

Overview of Software Development Processes





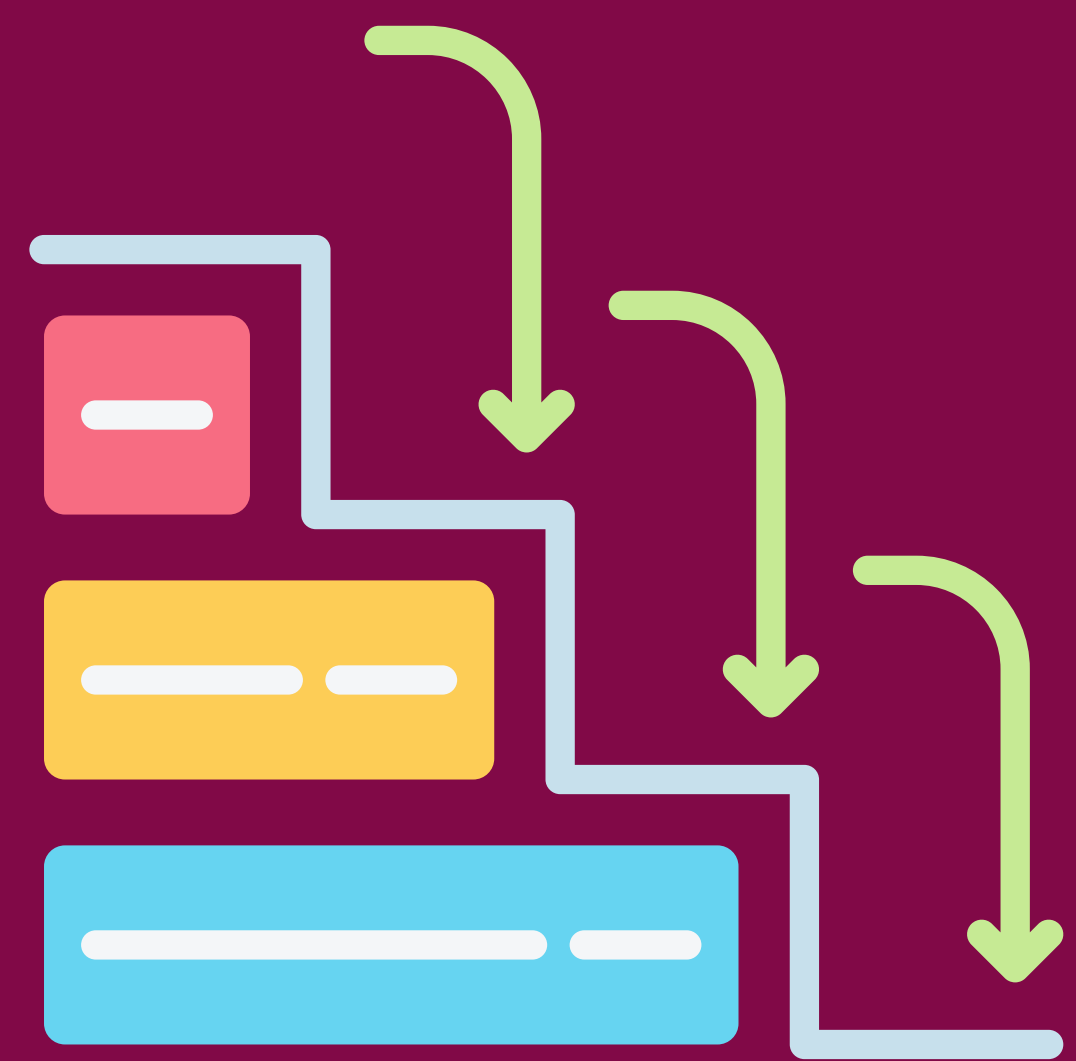
These are not all the software
development processes, and teams
often use variations of them.



Waterfall Model:

A linear and sequential approach where each phase must be completed before the next begins. Best for projects with well-defined requirements.

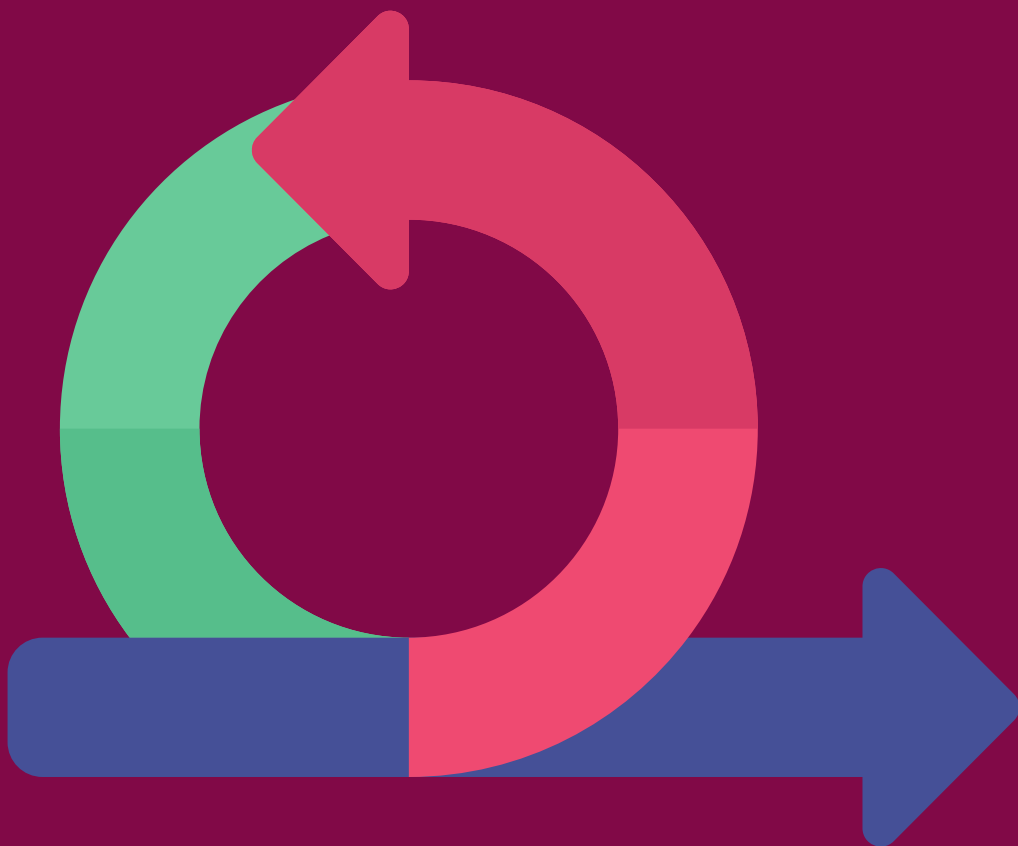
- Phases: Requirements, Design, Implementation, Testing, Deployment, Maintenance.



Agile Methodology:

An iterative and incremental approach focusing on flexibility and customer feedback. Ideal for projects with changing requirements.

- *Frameworks: Scrum, Kanban, Extreme Programming (XP).*





Agile Methodology



Scrum Framework:

- Roles : Product Owner, Scrum Master, Development Team.
- Artifacts : Product Backlog, Sprint Backlog, Increment.
- Ceremonies : Sprint Planning, Daily Standup, Sprint Review, Sprint Retrospective.

Agile Methodology Cont.

Kanban Framework:

- Principles: Visualize workflow, limit work in progress, enhance flow.
- Board: Columns representing different stages of the process (e.g., To Do, In Progress, Done).



To Do

In Progress

Testing

Done

Installer Update

Upgrade Production Server

Database Query Profiling

Improve Retention

Merge New Engine Build

Promotion Discount

Automatic Build

Upload Assets to Cloud

Dashboard KPI

Improve Server Response Times

Optimize Stats Queries

Switch to New Graph Format

New UI Icons

Server Load Balancing

Test New Performance Profiler

Merge Alpha 1 and Dev Branches

Integrate Predictive Analytics

AI Keyword Sorting

Revenue Dashboard

Test New Workstations

New Login Dialog

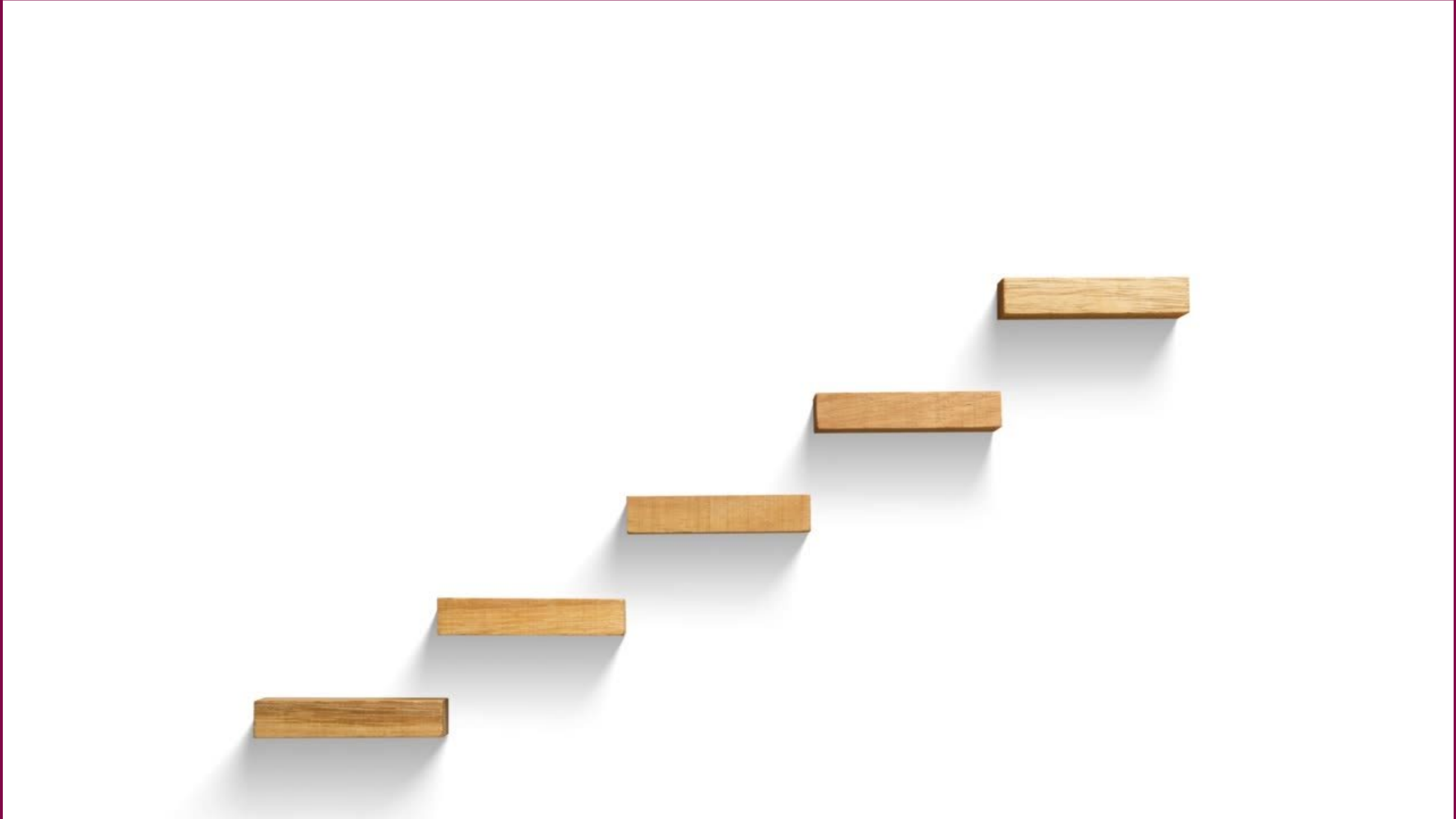
Upgrade Automatic Build System

Create Alpha 2 Branch

Optimize Loading

AI Lead Coder Job Desc

Export Item Shop List Feature



Other Development Processes

DevOps:

Combines development and operations to improve collaboration and productivity by automating workflows and continuously measuring application performance.

- Practices: Continuous Integration (CI), Continuous Deployment (CD), Infrastructure as Code (IaC).

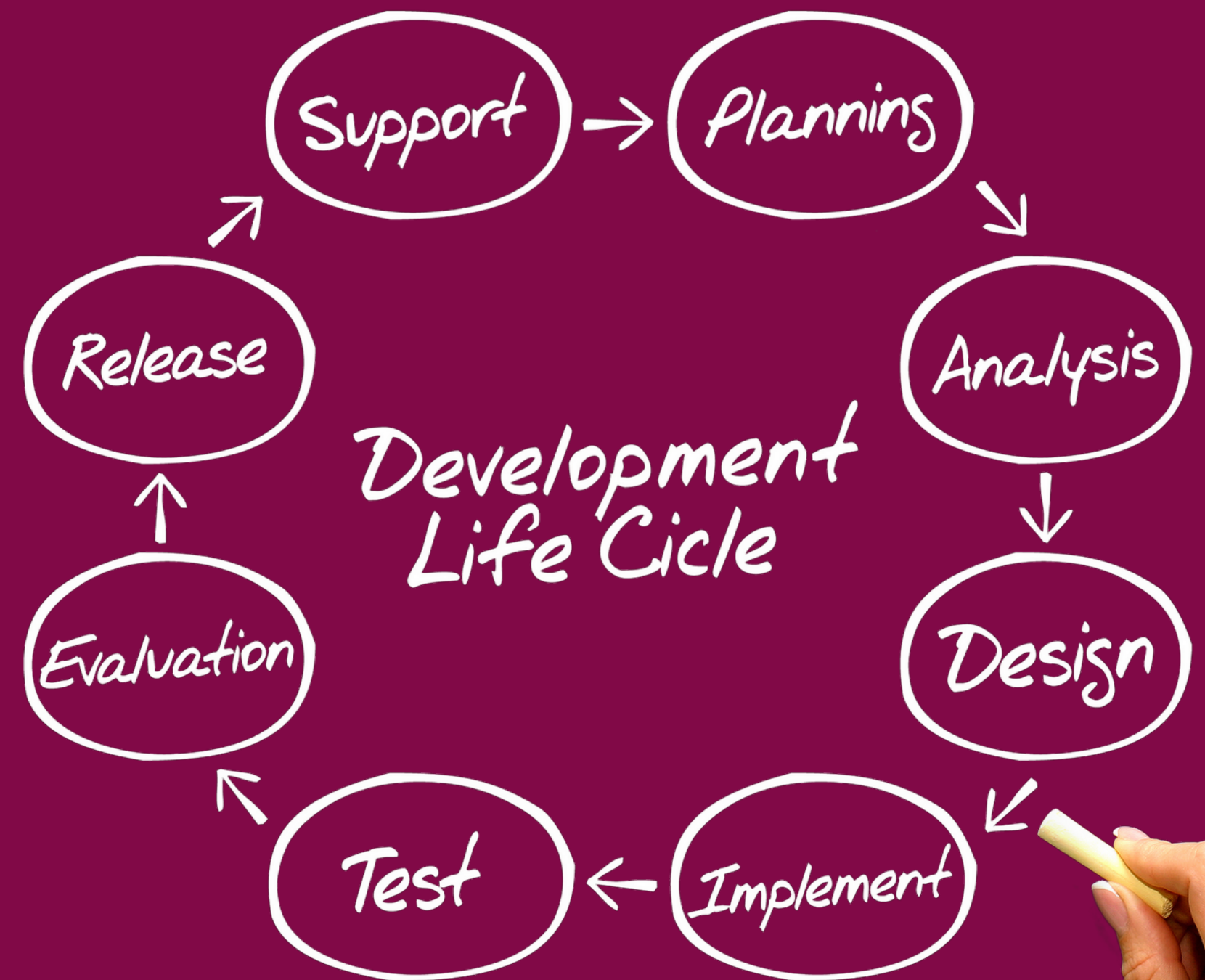
Other Development Processes Cont.

Lean Software Development:

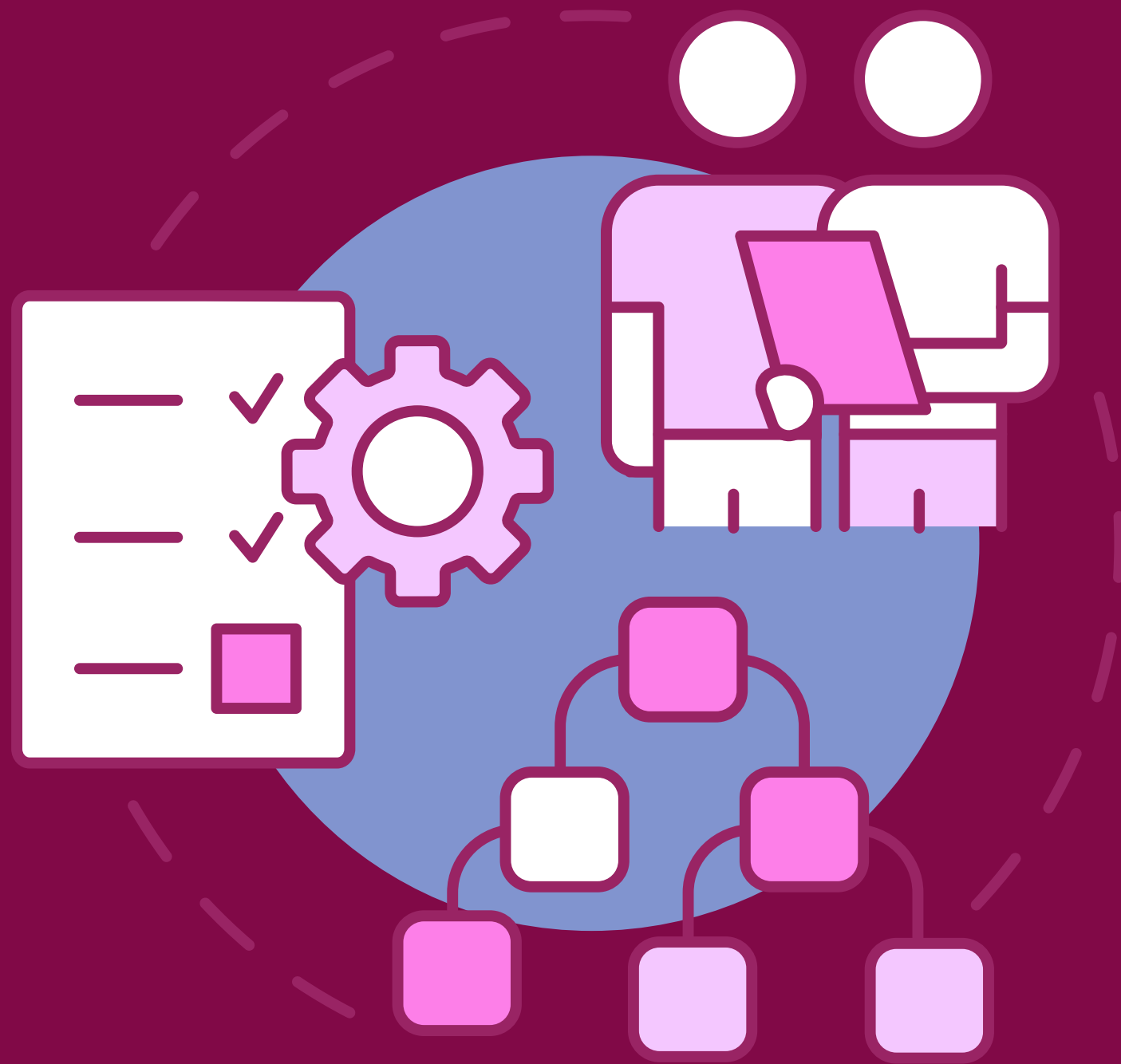
Focuses on minimizing waste and maximizing value delivery.

- Principles: Eliminate waste, Build quality in, Create knowledge, Defer commitment, Deliver fast, Respect people, Optimize the whole.

No matter what process you use, each step in the development cycle is essential. In Agile, these steps repeat every sprint. In Waterfall, they follow a sequential, step-by-step approach.



Requirements elicitation



- Gathers and documents user and business requirements.
- Ensures all stakeholders' needs are understood and addressed.

User Testing

- Involves real users to test the product.
- Collects feedback to improve usability and functionality.



Quality Assurance (QA)



- Tests the software to find and report bugs.
- Ensures the product meets quality standards.
- Works with developers to resolve issues.



Join the Discussion on Coursera

Discuss one software development process you have previously used & the benefits and challenges that came up.